



Observer-based fault estimation for a class of discrete-time switched affine systems: An application to the DC-DC converter

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Abstract

This paper studies the observer-based fault estimation for a class of discrete-time switched affine systems. A new class of fault estimator with an affine term is first established based on the state observer approach, and a target solution problem of the augmented error system is established under the H_∞ performance scheme. The min-type Lyapunov function and its associated switching function are used to obtain the stability condition of the augmented error system in terms of Lyapunov-Metzler inequality. The idea of minimizing the volume of the ellipsoid is adopted containing the set of attraction in order to obtain an attraction set as small as possible. Through solving an optimization problem, the stability of augmented error system can be ensured, and the corresponding fault estimator design is realized in the form of linear matrix inequalities. Finally, a practical application of three level DC-DC multicellular converter is illustrated to verify the effectiveness and practicability of the developed observer-based fault estimation approach.

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1. Introduction

As a special class of hybrid systems, switched systems consist of a finite number of subsystems and a switching rule responsible for activating one of them at each instant of time [1–4]. In particular the switched affine systems are a significant subclass characterised by the presence of affine terms, which can be used to model a large number of engineering systems in practice, especially in the field of power electronics [5,6]. The presence of affine terms is primarily due to the existence of multiple equilibrium points. As the affine terms are null, the corresponding system becomes a class of switched linear systems, that is only an unique equilibrium point exists. On the other hand, as the affine term is non-null, the corresponding issues of stability analysis, control and synthesis are more challenging. More recently, the related works have been appeared mainly focusing on the stability analysis and control issues. For instance, the problem of global asymptotic stability is addressed for a class of continuous-time switched affine systems [7,8]. The condition of quadratic stabilization has been obtained in [9], when a convex combination of subsystem matrices is Hurwitz, while another convex combination of affine vectors equals to zero, a switching law is designed to assure that the entire switched system is quadratically stable. In [10], the practical stabilization issue of switched affine systems has been addressed by the approach of hybrid systems. Based on time or space regularization, uniform dwell time is induced to get two solutions. In terms of ‘max’ composition approach, the global asymptotic stability of a class of switched affine system is analyzed in [11] by designing a switching rule. When the switching frequency is limited, the practical stability is assured in discrete-time domain [12]. In [13], the issues of stability analysis and control design have been involved for a class of switched affine systems, and the condition is derived by choosing a general quadratic Lyapunov function. Besides, some other works for stability and control synthesis also have been reported about the switched affine systems, see [14–16]. In addition, some literatures on chattering-free control or self-triggered formation control issue are represented in [17–22].

Along with the extension of the system scale, and the increasing of the complexity, it is of great significance to guarantee the reliability and security of complex systems [23,24]. Fault diagnosis and tolerant control technology has received more and more attention during the past three decades [25, 26, 27]. Fault diagnosis include fault detection, isolation and identification. First, determining whether the system has a fault, after fault detection, determining the fault type, the location of the fault, then determining the size and the occurring time of the fault. Generally, the active fault-tolerant control can be divided into three steps: fault detection and isolation, fault estimation and fault accommodation [28,29]. In particular, fault estimation provides the information of fault more abundantly, therefore, fault estimation plays a very important role to realize the fault tolerant control task. Reference [30] investigates the fault estimation problem of switched systems in the continuous-time domain under average dwell time switching signals, and a class of sliding mode observer design is proposed to reduce the observer dimensions. In [31], a switched descriptor state observer is established to estimate the faults information for a class of switched linear systems with time delays. Based on the hybrid Luenberger-observer method, the fault estimation issue has been studied for a class of piecewise affine systems in [32]. By using the disturbance decoupling idea, a proportional differential switched observer is developed in [33] to ensure the stability of error systems, and then the fault estimation issue is resolved for a class of singular switched systems with average dwell time switching property. To the best of the authors’ knowledge, turn to the switched affine systems, the fault estimation issue has not been fully studied and addressed, regardless

of discrete-time domain and continuous-time domain, which is one of main motivations for the current research.

In particular, some methods deserve particular attention for fault estimation issue, including the adaptive estimation, sliding mode observer and H_∞ approaches. For instance, the method based on adaptive observer [34] is mainly to establish the adaptive observer of the system and to achieve the performance of fault signal estimation by designing the adaptive fault estimation law. The main advantage of adaptive method that it can adjust the ability of processing data according to different data characteristics, such that the adaptive observation data and the actual output data can adapt to each other. However, the error system caused by the difference between the actual operating system and the prediction system under the state observer needs to meet the strict positive real condition. With the widespread application of sliding mode control in system models, fault estimation methods based on sliding mode observers have also been emerged recently. For instance, a classic sliding mode observer, combining with an equivalent output conversion method, is used to estimate and reconstruct the fault signal for a class of nonlinear system with uncertainties. Compared with other observer methods, the advantage of the sliding mode observer method is that it can handle the system with uncertainties very well and is robust to fault estimation. But when traditional sliding mode observer is applied to switched systems, the sliding mode surface tends to have the nature of modal dependence, so it is difficult to guarantee the strict accessibility of the sliding mode surface, which has certain limitations for the fault reconstruction method based on the equivalent output injection signal. Finally, the core advantage of H_∞ approach is to address the issue of robustness in the presence of disturbances and uncertainty. Based on the H_∞ approach, some results have been reported about the robust fault detection and fault estimation issues for switched linear systems in [35–39]. In [40], a robust fault detection filter design of mixed H_-/H_∞ approach based on multiple Lyapunov–Krasovskii function has been proposed for time-varying delays switched systems. A robust estimator with an integrated observer-based fault estimation and accommodation strategy has been investigated for discrete-time piecewise linear systems in [41]. Some other types of Lyapunov functions, for example, the input-to-state stable Lyapunov function is constructed to investigate the input-to-state stability (ISS) issue for nonlinear systems with stochastic impulses in [42]. In [43], a classis Halanay inequality is first extended to the time-varying sampled-data system, the problem of sampled-data control is investigated for a class of linear time-varying systems. One issue arises that general mathematical derivation methods based on H_∞ approach are incapable of using to obtain the condition of fault estimator design in terms of linear matrix inequality due to the existence of affine term for the switched affine system. The design methods of discrete switched linear systems in the above literature are incapable of using to extend to solve the issue of fault estimator design for discrete-time switched affine systems directly. Therefore, it is very necessary to develop a new method to deal with the fault estimation issue based on robust H_∞ method for switched affine systems, which is another motivation in this paper.

Motivated by the aforementioned discussions, the fault estimation issue is investigated for a class of discrete-time switched affine systems based on the Luenberger observer design approach in this paper. A novel min-type Lyapunov function, including its associated switching function, is constructed, which ensures the stability of augmented error systems. According to the Lyapunov–Metzler inequalities, new conditions about the stability of augmented error systems are established via the min-type Lyapunov function approach. In addition, after decoupling the disturbances through special scaling techniques, the min-type Lyapunov function approach and the idea of minimizing the volume of the ellipsoid containing the set of at-

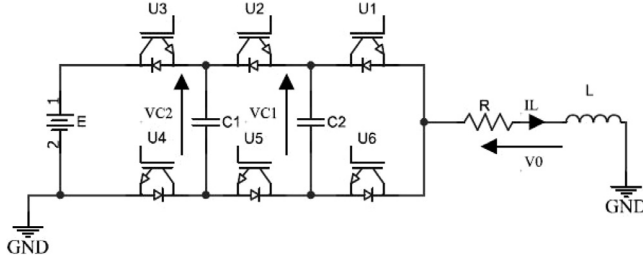


Fig. 1. The structure of a three-cell converter associated to an inductive load.

traction are extended to analyse the H_∞ performance. The guaranteed H_∞ performance index conditions based on Lyapunov-Metzler inequalities are obtained as the first attempt. Then the fault estimator design is carried out, which differs from the commonly used gain matrices design method. The main contributions in this paper are outlined as follows:

(1) Based on the framework of H_∞ observer, a class of Luenberger observer is considered to estimate the state and additive fault at the same time for a class of discrete-time switched affine systems as first attempt.

(2) The min-type Lyapunov function approach is applied to carry out the design of fault estimator for a class of discrete-time switched affine systems with state-dependent switching in a novel framework. The problem that is very difficult to handle due to the existence of the affine term in the presented switched affine system is resolved through the developed approach to obtain the conditions based on linear matrix inequalities technique.

(3) Efficient conditions of the robust fault estimation observer design for a class of discrete-time switched affine systems are obtained, and the simulation results demonstrate that the designed fault estimator can be used to perform the estimation of constant and even time-varying fault signals. The proposed fault estimator design method can be considered as a potential application of fault estimation for power electronics in the framework of switched affine systems.

Notations: The notation used in this paper is standard. For symmetric matrices, $(*)$ denotes each of these symmetric blocks. $\mathbb{K} = \{1, \dots, N\}$ is composed of N positive integers. $\mathbb{R}$ and $\mathbb{N}$ denote the sets of real and natural numbers, respectively. For any symmetric matrix, $X > 0$ denotes a positive definite matrix. $\text{zeros}(n)$ denotes all-zero matrix with n rows and n columns, and $\text{zeros}(m, n)$ denotes all-zero matrix with m rows and n columns. The unitary simplex is denoted as $\Lambda := \left\{ \lambda \in \mathbb{R}^N \mid \lambda_i \geq 0, \sum_{i=1}^N \lambda_i = 1, \forall i \in \mathbb{K} \right\}$.

2. Motivation example: three level multicellular DC-DC converter

Firstly, a multicellular DC-DC converter associated with an inductive load R-L is depicted in Fig. 1. The multicellular DC-DC converter structure is based on a series connection of commutation cells where each one is composed of pairs of complementary switches. Its state is controlled by a binary input signals $U_k \in \{0, 1\}$, $k = 1, 2, 3, 4, 5, 6$, where $U_1 = 1$ ($U_6 = 0$) indicates that the k th upper switch is closed (open) and k th lower switch is open (closed). By considering the capacitors voltages (V_{c1} and V_{c2}), and the load current (i_L) as the continuous states, the converter dynamics can be described by a third-order switched affine system with eight discrete modes. The discrete modes correspond to different input signals combinations

Table 1

The values of $U_1^{[\sigma]}$, $U_2^{[\sigma]}$, and $U_3^{[\sigma]}$, for each operation mode σ .

σ	1	2	3	4	5	6	7	8
$U_1^{[\sigma]}$	0	1	0	1	0	1	0	1
$U_2^{[\sigma]}$	0	0	1	1	0	0	1	1
$U_3^{[\sigma]}$	0	0	0	0	1	1	1	1

$(U_1^{[\sigma]}, U_2^{[\sigma]}, U_3^{[\sigma]})$ as follows: (0,0,0), (1,0,0), (0,1,0), (1,1,0), (0,0,1), (1,0,1), (0,1,1), (1,1,1), and the values of $U_1^{[\sigma]}$, $U_2^{[\sigma]}$, and $U_3^{[\sigma]}$ are given in Table 1 as a function of the current value of the switching signal σ . Namely, the converter presents eight operation modes (or subsystems) from which a switching signal σ selects one of them as active at each instant of time. For the dynamics of the three-cell DC-DC converter can be modelled as a class of continuous-time switched affine system [14,44].

$$\begin{cases} \dot{x}(t) = A_{c\sigma}x(t) + b_{c\sigma} + H_{1\sigma}d(t) \\ y(t) = C_{\sigma}x(t) + H_{2\sigma}d(t) \end{cases} \quad (1)$$

where the state $x(t) \triangleq [V_{c1} \ V_{c2} \ i_L]^T$, $d(t)$ is the exogenous disturbance. The system matrices are respectively represented as

$$A_{c\sigma} = \begin{bmatrix} 0 & 0 & \frac{U_2^{[\sigma]} - U_1^{[\sigma]}}{L} \\ 0 & 0 & \frac{U_3^{[\sigma]} - U_2^{[\sigma]}}{L} \\ \frac{U_1^{[\sigma]} - U_2^{[\sigma]}}{L} & \frac{U_2^{[\sigma]} - U_3^{[\sigma]}}{L} & -\frac{R}{L} \end{bmatrix}, b_{c\sigma} = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L}U_3^{[\sigma]} \end{bmatrix}, C_{\sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

As is well known, the switching power device is an indispensable part of various electrical equipment in modern society. The possible faults in power electronic circuits are generally divided into two types, i.e., one is a parametric fault [45] and the other is a structural fault [46]. In particular, the parametric fault refers to a fault that occurs when the parameter value of a circuit component drifts or changes away from the normal value. While a structural fault is that the fault occurs when a circuit power device open- or short-circuit.

3. Problem formulation and preliminaries

In this section, a class of discrete-time switched affine systems with actuator faults is introduced based on the three-level multicellular DC-DC converter model presented in Section II. A kind of actuator faults existing in the converter, for instance, the fault occurs in the power resource, is described in the following.

3.1. Fault model

Assuming that the sampling time is T , and the fault exists, a class of discrete-time switched affine systems can be established via discretizing the Eq. (1):

$$\begin{cases} x(k+1) = A_{\sigma}x(k) + D_{a\sigma}d(k) + E_{\sigma}f(k) + b_{\sigma} \\ y(k) = C_{\sigma}x(k) + D_{b\sigma}d(k) \end{cases} \quad (2)$$

Denoting $x(t) = x(k)$, $\forall k \in \mathbb{N}$, by using the Euler approach, the system matrices for discretized switched affine systems can be obtained as $A_{\sigma} = e^{A_{c\sigma}T}$, $b_{\sigma} = \int_0^T e^{A_{c\sigma}\tau} d\tau b_{c\sigma}$, $\forall \sigma \in \mathbb{K}$,

where $x(k) \in \mathbb{R}^n$ is the state, $b_\sigma \in \mathbb{R}^n$ is the affine term, T is the sampling period, $y(k) \in \mathbb{R}^m$ is the measurement output, $d(k) \in \mathbb{R}^p$ is the disturbance, $f(k) \in \mathbb{R}^q$ represents the fault, $d(k)$ belongs to $l_2[0, \infty)$, $\sigma : \mathbb{R}^n \rightarrow \mathbb{K}$ is the state-dependent switching function, it is responsible to select the subsystem which activates at each instant of time $k \in \mathbb{N}$.

3.2. Fault estimator design

The following observer-based fault estimator is constructed

$$\begin{cases} \hat{x}(k+1) = A_\sigma \hat{x}(k) + E_\sigma \hat{f}(k) + L_\sigma (y(k) - \hat{y}(k)) + b_\sigma \\ \hat{y}(k) = C_\sigma \hat{x}(k) \\ \hat{f}(k+1) = \hat{f}(k) + F_\sigma (y(k) - \hat{y}(k)) \end{cases} \quad (3)$$

where $\hat{x}(k) \in \mathbb{R}^n$ is the state estimation of system (2), $\hat{f}(k) \in \mathbb{R}^q$ is the fault estimation, $L_\sigma \in \mathbb{R}^{n \times m}$ and $F_\sigma \in \mathbb{R}^{q \times m}$ are the estimator gain matrices to be determined. As expressed in Eq. (3), the fault estimation system is also a class of switched affine systems, here we assume that it is the same as the switching signal of original system. Defining that the state estimation error $e_x(k) \triangleq x(k) - \hat{x}(k)$, fault estimation error $e_z(k) \triangleq f(k) - \hat{f}(k)$ and $\Delta f(k) \triangleq f(k+1) - f(k)$, and combining Eqs. (2) and (3), it gets

$$\begin{cases} x(k+1) = A_\sigma x(k) + D_{a\sigma} d(k) + E_\sigma f(k) + b_\sigma \\ e_x(k+1) = (A_\sigma - L_\sigma C_\sigma) e_x(k) + E_\sigma e_z(k) + (D_{a\sigma} - L_\sigma D_{b\sigma}) d(k) \\ e_z(k+1) = -F_\sigma C_\sigma e_x(k) + e_z(k) - F_\sigma D_{b\sigma} d(k) + \Delta f(k) \end{cases} \quad (4)$$

Then, denoting that the augmented state variable $\psi(k) \triangleq [x^T(k) \quad e_x^T(k) \quad e_z^T(k)]^T$ and the generalized disturbance $w(k) \triangleq [d^T(k) \quad \Delta f^T(k)]^T$, the augmented error system is given as follows:

$$\begin{cases} \psi(k+1) = \tilde{A}_\sigma \psi(k) + \tilde{B}_\sigma w(k) + \tilde{D}_\sigma \\ e_z(k) = \tilde{C}_{z\sigma} \psi(k) \end{cases} \quad (5)$$

where

$$\begin{aligned} \tilde{A}_\sigma &\triangleq \begin{bmatrix} A_\sigma & 0 & 0 \\ 0 & A_\sigma - L_\sigma C_\sigma & E_\sigma \\ 0 & -F_\sigma C_\sigma & I_q \end{bmatrix}, \tilde{B}_{\sigma(k)} \triangleq \begin{bmatrix} D_{a\sigma} & 0 \\ D_{a\sigma} - L_\sigma D_{b\sigma} & 0 \\ -F_\sigma D_{b\sigma} & I_q \end{bmatrix}, \\ \tilde{C}_{z\sigma} &\triangleq [0 \quad 0 \quad I_q], \tilde{D}_\sigma \triangleq \begin{bmatrix} b_\sigma \\ 0 \\ 0 \end{bmatrix}. \end{aligned}$$

Furthermore, defining $\zeta(k) \triangleq \psi(k) - \psi_d$, then we can obtain the switched affine error system

$$\begin{cases} \zeta(k+1) = \tilde{A}_\sigma \zeta(k) + \varphi_\sigma + \tilde{B}_\sigma w(k) \\ e_z(k) = \tilde{C}_{z\sigma} \zeta(k) \end{cases} \quad (6)$$

where $\psi_d \triangleq [x_d \quad 0 \quad 0]^T$ and $\varphi_\sigma \triangleq (\tilde{A}_\sigma - I) \psi_d + \tilde{D}_\sigma$, x_d is the equilibrium point, $e_z(k)$ is the fault estimation error. After transformation, the equilibrium point that we are interested in is the origin $\zeta = 0$. A state-dependent switching function $\sigma(\zeta)$ including a set of attraction χ is provided for switched affine systems (6) by using the following definition.

Definition 1. [12,13] The set $\chi \subset \mathbb{R}^n$ is a set of attraction of system (6) regulated by the switching function $\sigma(\zeta)$ if there exists a min-type Lyapunov function $V(\zeta)$, such that the following conditions are satisfied:

- (i) $0 \in \chi$,
- (ii) If $\zeta \notin \chi$ then $\Delta V(\zeta) < 0$.

Remark 1. In some cases, the Definition 1 splits up into the context of stability, which has been adopted to analyze the switched affine systems in discrete-time domain. In general, the switching function $\sigma(\zeta)$ needs to be designed to impose the state $\psi(k)$ towards a set of attraction χ , which contains the desired equilibrium point. The main objective is to minimize the volume of an ellipsoid that includes the set χ in its interior to get the set χ as small as possible.

In order to accomplish the above goal, a class of min-type Lyapunov function is borrowed

$$V(\zeta) = \min_{i \in \mathbb{K}} \zeta^T P_i \zeta \quad (7)$$

and its associated switching function is given in the following

$$\sigma(\zeta) = \arg \min_{i \in \mathbb{K}} \zeta^T P_i \zeta \quad (8)$$

The objective of this paper is to design the fault estimator Eq. (3), by using the min-type Lyapunov function $V(\zeta)$ and its associated switching function $\sigma(\zeta)$, such that:

(1) The error system (6) is asymptotically stable,

(2) Assuming that the initial state of error system (6) $\psi(k_0) \equiv 0$, the fault estimation error $e_z(k)$ is robust to $w(k)$, that is, there exist a scalar $\gamma > 0$, the following H_∞ performance constraint is satisfied:

$$\|e_z\|_2 < \gamma \|w\|_2.$$

4. Main results

In this section, the stability conditions of augmented switched affine error system are derived by means of min-type Lyapunov function and its associated switching function technique, and the corresponding desired performance requirement is satisfied. The gain matrices L_i, F_i are respectively designed to adequately deal with the fault estimation issue for the presented discrete-time switched affine system (2) under the state-dependent switching.

4.1. Stability analysis

Let us consider the augmented switched affine error system given by

$$\zeta(k+1) = \tilde{A}_\sigma \zeta(k) + \varphi_\sigma \quad (9)$$

for all $k \in \mathbb{N}$. In the following, the main goal is to preserve the stability of augmented switched affine error system (9) by using the Lyapunov-Metzler inequality technique. At first, let us recall the class of Metzler matrices denoted by $\mathcal{Q}$, and constituted by all matrices $\Pi \in \mathbb{R}^{\mathbb{K} \times \mathbb{K}}$ with elements π_{ji} , such that

$$\pi_{ji} \geq 0, \sum_{j=1}^{\mathbb{K}} \pi_{ji} = 1, \forall i, j. \quad (10)$$

It is clear that any $\Pi \in \mathcal{Q}$, where Π presents an unitary eigenvalue, which satisfies $c^T \Pi = c^T$ with $c^T = [1, \dots, 1]$. In terms of the Gershgorin Circle Theorem [47], all eigenvalues of Π are located inside circles centered at $\pi_{ii} \geq 0$ with radius $\sum_{j=1}^{\mathbb{K}} \pi_{ji} = 1 - \pi_{ii}$ and therefore, all eigenvalues are inside the unit circle. Before proceedings, let us define the matrix

$$P_{\pi_i} \triangleq \sum_{j=1}^{\mathbb{K}} \pi_{ji} P_j, \quad (11)$$

where $\pi_{ji} \geq 0$, $\sum_{j=1}^{\mathbb{K}} \pi_{ji} = 1$ and $P_j > 0$.

Consider the augmented switched affine error system (9), with the aid of the min-type Lyapunov function (7) and its associated switching function (8), the stability conditions for error system (9) can be derived in the following theorem.

Theorem 1. Consider the system (9), let ψ_d be given and assume that the estimator gains L_σ and F_σ in Eq. (3) are known. If there exist symmetric matrices M , P_i , a Metzler matrix $\Pi \in \mathcal{Q}$, a vector $\hbar$ of compatible dimensions and positive scalars α_i , satisfying the following optimization problem

$$\inf_{P_i, \Pi, M, \hbar, \alpha_i} -\ln(\det(M)), \quad (12)$$

subject to the matrix inequalities

$$\begin{bmatrix} \alpha_i P_i & 0 & M & \alpha_i \tilde{A}_i^T P_{\pi_i} \\ * & 1 & -\hbar^T & \alpha_i \varphi_i^T P_{\pi_i} \\ * & * & M & 0 \\ * & * & * & \alpha_i P_{\pi_i} \end{bmatrix} > 0, \quad (13)$$

then the switching function (8) ensures that the set centered at $\tau = M^{-1} \hbar$ can be given as

$$\epsilon = \{\zeta \in \mathbb{R}^{n_x} : (\zeta - \tau)^T M (\zeta - \tau) \leq 1\} \quad (14)$$

Proof. Consider the trajectory of system (9), assume that at an instant of time $k \in \mathbb{N}$, the switching rule is $\sigma = i \in \mathbb{K}$. To simplify the notation, we denote $\zeta(k) \triangleq \zeta$. The min-type Lyapunov function provides

$$\begin{aligned} \Delta V[\zeta(k)] &= V[\zeta(k+1)] - V[\zeta(k)] \\ &= \min \left(\tilde{A}_i \zeta + \varphi_i \right)^T P_\kappa \left(\tilde{A}_i \zeta + \varphi_i \right) - \zeta^T P_i \zeta \\ &= \min \left(\tilde{A}_i \zeta + \varphi_i \right)^T P_\lambda \left(\tilde{A}_i \zeta + \varphi_i \right) - \zeta^T P_i \zeta \\ &\leq \left(\tilde{A}_i \zeta + \varphi_i \right)^T P_{\pi_i} \left(\tilde{A}_i \zeta + \varphi_i \right) - \zeta^T P_i \zeta \\ &\leq \zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i \right) \zeta + 2 \zeta^T \tilde{A}_i^T P_{\pi_i} \varphi_i + \varphi_i^T P_{\pi_i} \varphi_i, \end{aligned} \quad (15)$$

Let $\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i = -\Omega_i$ and assuming that $\Omega_i > 0$ holds, combining Definition 1 and inequality Eq. (15), it has,

$$\zeta^T (P_i - \tilde{A}_i^T P_{\pi_i} \tilde{A}_i) \zeta - 2 \zeta^T \tilde{A}_i^T P_{\pi_i} \varphi_i - \varphi_i^T P_{\pi_i} \varphi_i \leq 0. \quad (16)$$

Then, rewrite inequality Eq. (16) in the form of set η_s with $\eta_s := \cup_{i \in \mathbb{K}} \epsilon_i$ as shown in the following:

$$\epsilon_i = \{ \zeta \in \mathbb{R}^{n_\zeta} : (\zeta - \tau_i)^T \Omega_i (\zeta - \tau_i) \leq a_i^T \Omega_i^{-1} a_i + d_i \},$$

where $\tau_i \triangleq \Omega_i^{-1} a_i$, $a_i \triangleq \tilde{A}_i^T P_{\pi i} \varphi_i$ and $d_i \triangleq \varphi_i^T P_{\pi i} \varphi_i$.

In the following, we want to find the smallest ellipsoid ϵ which contains η_s in its interior.

Firstly, the inequality Eq. (13) implies

$$\begin{bmatrix} \alpha_i P_i & \alpha_i \tilde{A}_i P_{\pi i} \\ * & \alpha_i P_{\pi i} \end{bmatrix} > 0,$$

namely,

$$\Omega_i = P_i - \tilde{A}_i^T P_{\pi i} \tilde{A}_i > 0, \forall i \in \mathbb{K},$$

is guaranteed.

Moreover, replacing $\tilde{h} = M\tau$ in Eq. (13), by using the Schur Complement with respect to inequality Eq. (13), it yields that:

$$\begin{bmatrix} \alpha_i P_i & 0 \\ * & 1 \end{bmatrix} + \begin{bmatrix} M \\ -\tilde{h}^T \end{bmatrix} (-M)^{-1} \begin{bmatrix} M & -\tilde{h} \end{bmatrix} + \alpha_i \begin{bmatrix} \tilde{A}_i^T P_{\pi i} \\ \varphi_i^T P_{\pi i} \end{bmatrix} (-P_{\pi i})^{-1} \begin{bmatrix} P_{\pi i} \tilde{A}_i & P_{\pi i} \varphi_i \end{bmatrix} > 0 \quad (17)$$

then, the inequality Eq. (17) can be rewritten into the following form:

$$\begin{bmatrix} M & -M\tau \\ * & \tau^T M\tau - 1 \end{bmatrix} < \alpha_i \begin{bmatrix} P_i - \tilde{A}_i^T P_{\pi i} \tilde{A}_i & -\tilde{A}_i^T P_{\pi i} \varphi_i \\ * & -\varphi_i^T P_{\pi i} \varphi_i \end{bmatrix}, i \in \mathbb{K}.$$

namely,

$$\begin{bmatrix} M & -M\tau \\ * & \tau^T M\tau - 1 \end{bmatrix} < \alpha_i \begin{bmatrix} \Omega_i & -a_i \\ * & -d_i \end{bmatrix}, i \in \mathbb{K}. \quad (18)$$

Next, the terms on the left side of inequality Eq. (18) are multiplied by $[\zeta^T \quad 1]^T$ to the right and its transpose to the left, respectively, it gets

$$(\zeta - \tau)^T M (\zeta - \tau) - 1 < \alpha_i [\zeta^T \Omega_i \zeta - a_i^T \zeta - \zeta^T a_i - d_i].$$

By the idea of S-procedure, the ellipsoid ϵ in Eq. (14) contains the union of sets ϵ_i . The objective function is used to minimize the volume of ϵ . The proof is completed. $\square$

To obtain simpler-to-solve condition, which is, dependent only on the parameters of the Metzler matrix, considering that $\alpha_i = \alpha$, $\forall i \in \mathbb{K}$, the following corollary is given.

Corollary 1. *The conclusion in Theorem 1 still holds whenever is replaced by*

$$\begin{bmatrix} P_i & 0 & M & \tilde{A}_i^T P_{\pi i} \\ * & 1 & -\tilde{h}^T & \varphi_i^T P_{\pi i} \\ * & * & M & 0 \\ * & * & * & P_{\pi i} \end{bmatrix} > 0, \quad (19)$$

for all $P_i > 0$.

Proof. The proof comes from Eq. (13) by designating $\alpha_i = \alpha > 0$ and redefining $P_i \rightarrow \alpha P_i$ and the detailed steps are omitted here. $\square$

Remark 2. Compared with the conditions of [Theorem 1](#), the conditions in [Corollary 1](#) are more easily operable numerically. We can assume that one of α_i is equal to one. This can be easily verified by substituting $P_i = S_i/\alpha_1$ and defining $\kappa_i = \alpha_i/\alpha_1$ in (13), we represent $P_i \rightarrow S_i$ and $\alpha_i \rightarrow \kappa_i$, and the choice will not affect the result and the developments afterwards [\[12\]](#). This demonstrates that we can fix $\alpha_1 = 1$ in (13) and let the scalars $\alpha_i > 0, \forall i \in \mathbb{K}$.

Remark 3. The condition implied in [Theorem 1](#) and [Corollary 1](#) is provided as follows:

$$\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i < 0, \forall i \in \mathbb{K}. \quad (20)$$

It contains the inequality

$$\sum_{i \in \mathbb{K}} \lambda_i \tilde{A}_i^T S \tilde{A}_i - S < 0, \quad (21)$$

with $S > 0$ and $\lambda_i \in \Lambda$ as a particular case. In [\[13,48\]](#), the inequality [Eq. \(21\)](#) has been verified as a condition, which ensures the existence of the desired attractive set.

Remark 4. The products of variables π_{ji} and P_j of condition [Eq. \(19\)](#) specify the bilinear matrix inequalities, which is a NP-hard problem. One alternative algorithm is to conduct a multi-dimensional search on π_{ji} and through specifying π_{ji} to solve the corresponding linear matrix inequalities in [\[49\]](#), or by setting the same diagonal elements of Metzler matrix, i.e., $\pi_{ii} = \rho$ and $\sum_{j=1, j \neq i}^N \pi_{ji} = 1 - \rho$, with the aid of some solvers such as YALMIP [\[50\]](#).

4.2. Performance analysis

Through using the min-type Lyapunov function [Eq. \(7\)](#) and its associated switching function [Eq. \(8\)](#), the corresponding conditions satisfying the desired performance requirement are obtained in the following theorem.

Theorem 2. Consider the augmented error system [Eq. \(6\)](#), let $\psi_d, \delta_1 > 0, \delta_2 > 0, \alpha_i > 0$ be given, where δ_1, δ_2 and α_i are known constants. If there exist symmetric matrices M, P_i , a Metzler matrix $\Pi \in \mathcal{Q}$, $\tilde{A}_i$, a vector h of compatible dimensions, the following optimization problem is satisfied

$$\inf_{P_i, \Pi, M, h, \tilde{A}_i} -\ln(\det(M)) \quad (22)$$

subject to

$$\begin{bmatrix} \Omega_{11} & \Omega_{12} \\ * & \Omega_{22} \end{bmatrix} > 0 \quad (23)$$

$$\begin{bmatrix} \Upsilon_{11,s} & \Upsilon_{12,s} \\ * & \Upsilon_{22,s} \end{bmatrix} > 0 \quad (24)$$

where

$$\Omega_{11} \triangleq \begin{bmatrix} \alpha_i P_i & 0 \\ 0 & 1 \end{bmatrix}, \quad \Omega_{12} \triangleq [\Omega_{12,s} \quad \Omega_{21,s}]$$

$$\Omega_{12,s} \triangleq \begin{bmatrix} M & \alpha_i \sqrt{\delta_1} \tilde{A}_i^T P_{\pi_i} & 0 \\ -h^T & 0 & \alpha_i \sqrt{\delta_2} \varphi_i^T P_{\pi_i} \end{bmatrix}, \quad \Omega_{21,s} \triangleq \begin{bmatrix} \alpha_i \tilde{A}_i^T P_{\pi_i} & \alpha_i C_{zi}^T \\ \alpha_i \varphi_i^T P_{\pi_i} & 0 \end{bmatrix}$$

$$\Omega_{22} \triangleq \text{diag}\{M, \alpha_i P_{\pi_i}, \alpha_i P_{\pi_i}, \alpha_i P_{\pi_i}, \alpha_i I_q\}, \quad \Upsilon_{11,s} \triangleq \gamma^2 I$$

$$\Upsilon_{12,s} \triangleq [\tilde{B}_i^T P_{\pi_i} \quad \delta_1^{-1} \tilde{B}_i^T P_{\pi_i} \quad \delta_2^{-1} \tilde{B}_i^T P_{\pi_i}], \quad \Upsilon_{22,s} \triangleq \text{diag}\{P_{\pi_i}, \delta_1^{-1} P_{\pi_i}, \delta_2^{-1} P_{\pi_i}\}$$

then the switching function Eq. (8) ensures that the set centered at $\phi = M^{-1}h$ is given as

$$\nu = \{\zeta \in \mathbb{R}^{n_x} : (\zeta - \phi)^T M(\zeta - \phi) \leq 1\}. \quad (25)$$

Proof. Consider the trajectory of system (6), assume that the switching rule is $\sigma = i \in \mathbb{K}$ at an instant of time $k \in \mathbb{N}$. To simplify the notation, we denote $\zeta(k) \triangleq \zeta$. The min-type Lyapunov function provides

$$\begin{aligned} \Delta V[\zeta(k)] &= V[\zeta(k+1)] - V[\zeta(k)] \\ &= \min \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right)^T P_{\kappa} \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right) - \zeta^T P_i \zeta \\ &\quad + e_z^T e_z - \gamma^2 w^T w - e_z^T e_z + \gamma^2 w^T w \\ &= \min \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right)^T P_{\lambda} \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right) - \zeta^T P_i \zeta \\ &\quad + \zeta^T \tilde{C}_{zi}^T \tilde{C}_{zi} \zeta - \gamma^2 w^T w - e_z^T e_z + \gamma^2 w^T w \\ &\leq \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right)^T P_{\pi_i} \left(\tilde{A}_i \zeta + \varphi_i + \tilde{B}_i w \right) - \zeta^T P_i \zeta \\ &\quad + \zeta^T \tilde{C}_{zi}^T \tilde{C}_{zi} \zeta - \gamma^2 w^T w - e_z^T e_z + \gamma^2 w^T w \\ &\leq \zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i + \tilde{C}_{zi}^T \tilde{C}_{zi} \right) \zeta + 2\zeta^T \tilde{A}_i^T P_{\pi_i} \varphi_i \\ &\quad + 2\zeta^T \tilde{A}_i^T P_{\pi_i} \tilde{B}_i w + 2\varphi_i^T P_{\pi_i} \tilde{B}_i w + w^T \left(\tilde{B}_i^T P_{\pi_i} \tilde{B}_i - \gamma^2 I \right) w \\ &\quad + \varphi_i^T P_{\pi_i} \varphi_i - e_z^T e_z + \gamma^2 w^T w \\ &\leq \zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i + \tilde{C}_{fi}^T \tilde{C}_{fi} \right) \zeta + 2\zeta^T \tilde{A}_i^T P_{\pi_i} \varphi_i \\ &\quad + \delta_1 \zeta^T \tilde{A}_i^T P_{\pi_i} \tilde{A}_i \zeta + \delta_1^{-1} w^T \tilde{B}_i^T P_{\pi_i} \tilde{B}_i w + \delta_2 \varphi_i^T P_{\pi_i} \varphi_i \\ &\quad + \delta_2^{-1} w^T \tilde{B}_i^T P_{\pi_i} \tilde{B}_i w + w^T \left(\tilde{B}_i^T P_{\pi_i} \tilde{B}_i - \gamma^2 I \right) w \\ &\quad + \varphi_i^T P_{\pi_i} \varphi_i - e_z^T e_z + \gamma^2 w^T w \\ &= \zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i + \tilde{C}_{zi}^T \tilde{C}_{zi} + \delta_1 \tilde{A}_i^T P_{\pi_i} \tilde{A}_i \right) \zeta \\ &\quad + 2\zeta^T a_i + f_i + w^T \left(\tilde{B}_i^T P_{\pi_i} \tilde{B}_i - \gamma^2 I + \delta_1^{-1} \tilde{B}_i^T P_{\pi_i} \tilde{B}_i + \delta_2^{-1} \tilde{B}_i^T P_{\pi_i} \tilde{B}_i \right) w \\ &\quad - e_z^T e_z + \gamma^2 w^T w, \end{aligned} \quad (26)$$

where $a_i \triangleq \tilde{A}_i^T P_{\pi_i} \varphi_i$, $f_i \triangleq (1 + \delta_2) \varphi_i^T P_{\pi_i} \varphi_i$.

If the following inequality is satisfied

$$\tilde{B}_i^T P_{\pi_i} \tilde{B}_i - \gamma^2 I + \delta_1^{-1} \tilde{B}_i^T P_{\pi_i} \tilde{B}_i + \delta_2^{-1} \tilde{B}_i^T P_{\pi_i} \tilde{B}_i < 0,$$

namely, Eq. (24) holds, at the same time,

$$\zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i + \tilde{C}_{zi}^T \tilde{C}_{zi} + \delta_1 \tilde{A}_i^T P_{\pi_i} \tilde{A}_i \right) \zeta + 2\zeta^T a_i + f_i < 0$$

holds, then

$$\Delta V[\zeta(k)] \leq \zeta^T \left(\tilde{A}_i^T P_{\pi_i} \tilde{A}_i - P_i + \tilde{C}_{zi}^T \tilde{C}_{zi} + \delta_1 \tilde{A}_i^T P_{\pi_i} \tilde{A}_i \right) \zeta + 2\zeta^T a_i + f_i$$

$$\begin{aligned}
 & +w^T \left(\tilde{B}_i^T P_{\pi i} \tilde{B}_i - \gamma^2 I + \delta_1^{-1} \tilde{B}_i^T P_{\pi i} \tilde{B}_i + \delta_2^{-1} \tilde{B}_i^T P_{\pi i} \tilde{B}_i \right) w - e_z^T e_z + \gamma^2 w^T w \\
 & \leq -e_z^T e_z + \gamma^2 w^T w.
 \end{aligned} \tag{27}$$

Next, we let $\tilde{A}_i^T P_{\pi i} \tilde{A}_i - P_i + \tilde{C}_{zi}^T \tilde{C}_{zi} + \delta_1 \tilde{A}_i^T P_{\pi i} \tilde{A}_i = -\vartheta_i$ and it makes that $\vartheta_i > 0$ holds.

Define the set $\mathcal{Q}_s$ as $\mathcal{Q}_s := \cup_{i \in \mathbb{K}} \mathcal{V}_i$ with

$$\mathcal{V}_i = \{ \zeta \in R^{n_x} : (\zeta - \phi_i)^T \vartheta_i (\zeta - \phi_i) \leq a_i^T \vartheta_i^{-1} a_i + f_i \}, \tag{28}$$

where $\phi_i \triangleq \vartheta_i^{-1} a_i$.

On the other hand, the inequality Eq. (23) implies

$$\vartheta_i = P_i - \tilde{A}_i^T P_{\pi i} \tilde{A}_i - \tilde{C}_{zi}^T \tilde{C}_{zi} - \delta_1 \tilde{A}_i^T P_{\pi i} \tilde{A}_i > 0, \forall i \in \mathbb{K}.$$

In the following, we want to find the smallest ellipsoid $\mathcal{V}$ which contains $\mathcal{Q}_s$ in its interior.

Firstly, replacing $h = M\phi$ in Eq. (23), by using the Schur Complement with respect to inequality Eq. (23), it yields that:

$$\begin{bmatrix} M & -M\phi \\ * & \phi^T M\phi - 1 \end{bmatrix} < \alpha_i \begin{bmatrix} \vartheta_i & -a_i \\ * & -f_i \end{bmatrix}, i \in \mathbb{K},$$

then multiplying both sides to the right by $[\zeta^T \ 1]^T$ and to the left by its transpose, it obtains

$$(\zeta - \phi)^T M (\zeta - \phi) - 1 < \alpha_i [\zeta^T \vartheta_i \zeta - a_i^T \zeta - \zeta^T a_i - f_i].$$

By using S-procedure, the ellipsoid ϵ in Eq. (25) contains the union of sets $\mathcal{V}$. The objective function is used to minimize the volume of $\mathcal{V}$.

Summing both sides of the inequality Eq. (27) from $k = 0$ to the infinity, and recalling that $\lim_{k \rightarrow \infty} V[\zeta(k)] = 0$ as a consequence of the asymptotic stability of the origin $\zeta = 0$, we obtain

$$\|e_z\|_2^2 - \gamma \|w\|_2^2 < 0, \tag{29}$$

which assures the performance index requirement is satisfied. The proof is completed. $\square$

Remark 5. From the second inequality to the third inequality of inequality Eq. (26), the decoupling process is performed, that is, $a^T P b + b^T P a \leq \delta a^T P a + \delta^{-1} b^T P b$. Therefore, it gets $2\zeta^T \tilde{A}_i^T P_{\pi i} \tilde{B}_i w \leq \delta_1 \zeta^T \tilde{A}_i^T P_{\pi i} \tilde{A}_i \zeta + \delta_1^{-1} w^T \tilde{B}_i^T P_{\pi i} \tilde{B}_i w$ and $2\varphi_i^T P_{\pi i} \tilde{B}_i w \leq \delta_2 \varphi_i^T P_{\pi i} \varphi_i + \delta_2^{-1} w^T \tilde{B}_i^T P_{\pi i} \tilde{B}_i w$. By using this scaling operation, the technique developed in Theorem 1 can be used well in the proof of Theorem 2.

4.3. Observer design

Theorem 3. Consider the augmented error system (6), let ψ_d , $\delta_1 > 0$, $\delta_2 > 0$, $\alpha_i > 0$ be given, where δ_1 , δ_2 and α_i are known constants. If there exist symmetric matrices M_1 , M_2 , M_3 , P_{it} ($i \in \mathbb{K}$, $t = 1, 2, 3$), $P_{\pi it}$ ($i \in \mathbb{K}$, $t = 1, 2, 3$) and matrices X_i , Y_i , vectors h_1 , h_2 , h_3 of compatible dimensions, such that the following optimization problem holds

$$\inf_{P_i, \Pi, M, h, \tilde{A}_i} -\ln(\det(M)) \tag{30}$$

subject to

$$\begin{bmatrix} \Phi_1 & \Gamma_1^T \\ \Gamma_1 & \Lambda_1 \end{bmatrix} > 0 \quad (31)$$

$$\begin{bmatrix} \Xi_1 & \Theta_1^T \\ \Theta_1 & \Upsilon_1 \end{bmatrix} > 0. \quad (32)$$

Then the fault estimator gains can be calculated as

$$L_i = P_{\pi i 2}^{-1} X_i, \quad F_i = P_{\pi i 3}^{-1} Y_i, \quad (33)$$

where

$$\begin{aligned} \Phi_1 &\triangleq \begin{bmatrix} \Phi_{1,1} & \Phi_{1,2} \\ \Phi_{2,1} & \Phi_{2,2} \end{bmatrix}, \quad \Gamma_1 \triangleq \begin{bmatrix} \Gamma_{1,1} & \Gamma_{1,2} \\ \Gamma_{2,1} & \Gamma_{2,2} \end{bmatrix}, \quad \Lambda_1 \triangleq \begin{bmatrix} \Lambda_{1,1} & 0 \\ 0 & \Lambda_{2,2} \end{bmatrix}, \\ \Xi_1 &\triangleq \begin{bmatrix} \Xi_{1,1} & \Xi_{1,2} \\ \Xi_{2,1} & \Xi_{2,2} \end{bmatrix}, \quad \Theta_1 \triangleq \begin{bmatrix} \Theta_{1,1} & \Theta_{1,2} \\ \Theta_{2,1} & \Theta_{2,2} \end{bmatrix}, \quad \Upsilon_1 \triangleq \begin{bmatrix} \Upsilon_{1,1} & 0 \\ 0 & \Upsilon_{2,2} \end{bmatrix}, \\ \Phi_{1,1} &\triangleq \text{diag}\{P_{i1}, P_{i2}, P_{i3}\}, \quad \Phi_{1,2} \triangleq \text{zeros}(3, 4), \quad \Phi_{2,1} \triangleq \begin{bmatrix} \Phi_{31,s} \\ \Phi_{32,s} \end{bmatrix}, \\ \Phi_{31,s} &\triangleq \begin{bmatrix} 0 & 0 & 0 \\ M_1 & 0 & 0 \end{bmatrix}, \quad \Phi_{32,s} \triangleq \begin{bmatrix} 0 & M_2 & 0 \\ 0 & 0 & M_3 \end{bmatrix}, \quad \Phi_{2,2} \triangleq \begin{bmatrix} \Phi_{11,s} & \Phi_{12,s} \\ \Phi_{21,s} & \Phi_{22,s} \end{bmatrix}, \\ \Phi_{11,s} &\triangleq \begin{bmatrix} 1 & 0 \\ -h_1 & M_1 \end{bmatrix}, \quad \Phi_{12,s} \triangleq \text{zeros}(2, 2), \quad \Phi_{21,s} \triangleq \begin{bmatrix} -h_2 & 0 \\ -h_3 & 0 \end{bmatrix}, \quad \Phi_{22,s} \triangleq \begin{bmatrix} M_2 & 0 \\ 0 & M_3 \end{bmatrix}, \\ \Gamma_{1,1} &\triangleq \begin{bmatrix} P_{\pi i 1} A_i & 0 & 0 \\ 0 & P_{\pi i 2} A_i - X_i C_i & P_{\pi i 2} E_i \\ 0 & -Y_i C_i & P_{\pi i 3} \end{bmatrix}, \quad \Gamma_{1,2} \triangleq \text{zeros}(3, 4), \quad \Gamma_{2,1} \triangleq \begin{bmatrix} \Gamma_{31,s} \\ \Gamma_{32,s} \end{bmatrix}, \\ \Gamma_{31,s} &\triangleq \text{zeros}(3), \quad \Gamma_{32,s} \triangleq \begin{bmatrix} P_{\pi i 1} A_i & 0 & 0 \\ 0 & P_{\pi i 2} A_i - X_i C_i & P_{\pi i 2} E_i \\ 0 & -Y_i C_i & P_{\pi i 3} \end{bmatrix}, \quad \Gamma_{2,2} \triangleq \begin{bmatrix} \Gamma_{11,s} & \Gamma_{12,s} \\ \Gamma_{21,s} & \Gamma_{22,s} \end{bmatrix}, \\ \Gamma_{11,s} &\triangleq \begin{bmatrix} P_{\pi i 1} \varphi_i & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \Gamma_{12,s} \triangleq \text{zeros}(3, 2), \quad \Gamma_{21,s} \triangleq \begin{bmatrix} P_{\pi i 1} \varphi_i & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \Gamma_{22,s} \triangleq \text{zeros}(3, 2), \\ \Lambda_{1,1} &\triangleq \text{diag}\{P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}, P_{\pi i 1}, P_{\pi i 2}\}, \quad \Lambda_{2,2} \triangleq \text{diag}\{P_{\pi i 3}, P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}\}, \\ \Xi_{1,1} &\triangleq \begin{bmatrix} \gamma^2 I & 0 \\ 0 & \gamma^2 I \end{bmatrix}, \quad \Xi_{1,2} \triangleq \text{zeros}(2, 3), \quad \Xi_{2,1} \triangleq \begin{bmatrix} P_{\pi i 1} D_{ai} & 0 \\ P_{\pi i 2} D_{ai} - X_i D_{bi} & 0 \\ -Y_i D_{bi} & P_{\pi i 3} \end{bmatrix}, \\ \Xi_{2,2} &\triangleq \text{diag}\{P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}\}, \quad \Theta_{1,1} \triangleq \begin{bmatrix} P_{\pi i 1} D_{ai} & 0 \\ P_{\pi i 2} D_{ai} - X_i D_{bi} & 0 \\ -Y_i D_{bi} & P_{\pi i 3} \end{bmatrix}, \\ \Theta_{1,2} &\triangleq \text{zeros}(3), \quad \Theta_{2,1} \triangleq \begin{bmatrix} P_{\pi i 1} D_{ai} & 0 \\ P_{\pi i 2} D_{ai} - X_i D_{bi} & 0 \\ -Y_i D_{bi} & P_{\pi i 3} \end{bmatrix}, \quad \Theta_{2,2} \triangleq \text{zeros}(3), \\ \Upsilon_{1,1} &= \text{diag}\{P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}\}, \quad \Upsilon_{2,2} = \text{diag}\{P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}\}, \end{aligned}$$

Proof. Define

$$P_i \triangleq \text{diag}\{P_{i1}, P_{i2}, P_{i3}\}, \quad P_{\pi i} \triangleq \text{diag}\{P_{\pi i 1}, P_{\pi i 2}, P_{\pi i 3}\},$$

$$M_i \triangleq \text{diag}\{M_1, M_2, M_3\}, h \triangleq [h_1 \quad h_2 \quad h_3]^T.$$

Put them into inequalities Eqs. (23) and (24), respectively to get

$$\begin{bmatrix} \Phi_1 & T_1^T \\ T_1 & \Lambda_1 \end{bmatrix} > 0 \quad (34)$$

$$\begin{bmatrix} \Sigma_1 & G_1^T \\ G_1 & \Upsilon_1 \end{bmatrix} > 0 \quad (35)$$

where

$$\begin{aligned} T_1 &\triangleq \begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix}, \quad T_{1,1} \triangleq \begin{bmatrix} P_{\pi i1}A_i & 0 & 0 \\ 0 & P_{\pi i2}A_i - P_{\pi i2}L_iC_i & P_{\pi i2}E_i \\ 0 & -P_{\pi i3}F_iC_i & P_{\pi i3} \end{bmatrix}, \\ T_{1,2} &\triangleq \text{zeros}(3, 4), \quad T_{2,1} \triangleq \begin{bmatrix} T_{31,s} \\ T_{32,s} \end{bmatrix}, \quad T_{31,s} \triangleq \text{zeros}(3), \\ T_{32,s} &\triangleq \begin{bmatrix} P_{\pi i1}A_i & 0 & 0 \\ 0 & P_{\pi i2}A_i - P_{\pi i2}L_iC_i & P_{\pi i2}E_i \\ 0 & -P_{\pi i3}F_iC_i & P_{\pi i3} \end{bmatrix}, \quad T_{2,2} \triangleq \begin{bmatrix} T_{11,s} & T_{12,s} \\ T_{21,s} & T_{22,s} \end{bmatrix}, \\ T_{11,s} &\triangleq \begin{bmatrix} P_{\pi i1}\varphi_i & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad T_{12,s} \triangleq \text{zeros}(3, 2), \quad T_{21,s} \triangleq \begin{bmatrix} P_{\pi i1}\varphi_i & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \\ T_{22,s} &\triangleq \text{zeros}(3, 2), \quad \Sigma_1 \triangleq \begin{bmatrix} \Sigma_{1,1} & \Sigma_{1,2} \\ \Sigma_{2,1} & \Sigma_{2,2} \end{bmatrix}, \quad \Sigma_{1,1} \triangleq \begin{bmatrix} \gamma^2 I & 0 \\ 0 & \gamma^2 I \end{bmatrix}, \\ \Sigma_{1,2} &\triangleq \text{zeros}(2, 3), \quad \Sigma_{2,1} \triangleq \begin{bmatrix} P_{\pi i1}D_{ai} & 0 \\ P_{\pi i2}D_{ai} - P_{\pi i2}L_iD_{bi} & 0 \\ -P_{\pi i3}F_iD_{bi} & P_{\pi i3} \end{bmatrix}, \\ \Sigma_{2,2} &\triangleq \text{diag}\{P_{\pi i1}, P_{\pi i2}, P_{\pi i3}\}, \quad G_1 \triangleq \begin{bmatrix} G_{1,1} & G_{1,2} \\ G_{2,1} & G_{2,2} \end{bmatrix}, \\ G_{1,1} &\triangleq \begin{bmatrix} P_{\pi i1}D_{ai} & 0 \\ P_{\pi i2}D_{ai} - P_{\pi i2}L_iD_{bi} & 0 \\ -P_{\pi i3}F_iD_{bi} & P_{\pi i3} \end{bmatrix}, \quad G_{1,2} \triangleq \text{zeros}(3), \\ G_{2,1} &\triangleq \begin{bmatrix} P_{\pi i1}D_{ai} & 0 \\ P_{\pi i2}D_{ai} - P_{\pi i2}L_iD_{bi} & 0 \\ -P_{\pi i3}F_iD_{bi} & P_{\pi i3} \end{bmatrix}, \quad G_{2,2} \triangleq \text{zeros}(3), \end{aligned}$$

then define $X_i \triangleq P_{\pi i2}L_i$, $Y_i \triangleq P_{\pi i3}F_i$, the expressions T_1 , Σ_1 and G_1 can be transformed into Γ_1 , Ξ_1 and Θ_1 , so the inequalities Eqs. (31) and (32) can be derived. The proof is completed. $\square$

Remark 6. The min-type Lyapunov function approach and the idea of minimizing the volume of the ellipsoid containing the set of attraction are used to analyse the H_∞ performance. The guaranteed H_∞ performance index is derived in Theorem 2 for a class of discrete-time switched affine systems as the first attempt. Its solving process is similar to the acquisition of the stability conditions of the error switched system, but in this process, the above ideas can be used to obtain the guaranteed H_∞ performance skillfully after a special solving process with a scaling technique, which is different from the traditional verification method of H_∞ performance index.

5. Illustrative examples

In this section, a class of multicellular DC-DC converter system is borrowed here, which has been discussed in Section II as a motivation to demonstrate the advantages and effectiveness of developed fault estimator design approach. The topology as shown in Fig. 1 presents $N = 8$ operation modes or subsystems. The values for U_1 , U_2 , and U_3 are given in Table 1 as a function of the current value of the switching signal σ . The system parameters are chosen as $V_{dc} = 60[V]$, $C_1 = C_2 = 40[\mu F]$, $L = 5[mH]$, and $R = 20[\Omega]$. The multicellular DC-DC converter system can be modelled as the continuous-time switched affine system (1). According to the Example 2 in [14], the system matrices are expressed as follows.

$$\begin{aligned} A_{c1} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix}, A_{c2} = \begin{bmatrix} 0 & 0 & -0.0125 \\ 0 & 0 & 0 \\ 0.1 & 0 & -2 \end{bmatrix}, \\ A_{c3} &= \begin{bmatrix} 0 & 0 & 0.0125 \\ 0 & 0 & -0.0125 \\ -0.1 & 0.1 & -2 \end{bmatrix}, A_{c4} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -0.0125 \\ 0 & 0.1 & -2 \end{bmatrix}, \\ A_{c5} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0.0125 \\ 0 & -0.1 & -2 \end{bmatrix}, A_{c6} = \begin{bmatrix} 0 & 0 & -0.0125 \\ 0 & 0 & 0.0125 \\ 0.1 & -0.1 & -2 \end{bmatrix}, \\ A_{c7} &= \begin{bmatrix} 0 & 0 & 0.0125 \\ 0 & 0 & 0 \\ -0.1 & 0 & -2 \end{bmatrix}, A_{c8} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \\ b_{c1} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, b_{c2} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, b_{c3} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, b_{c4} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \\ b_{c5} &= \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}, b_{c6} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}, b_{c7} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}, b_{c8} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

The sampling period is taken as $T = 0.5 [s]$, for all $t \in [t_n, t_{n+1})$, where $t_n = nT$ represents n -th sampling instant. Denoting $x(t_n) = x(k)$, $\forall k \in \mathbb{N}$, the system matrices for discretized switched affine systems can be obtained as $A_\sigma = e^{A_{c\sigma}T}$, $b_\sigma = \int_0^T e^{A_{c\sigma}\tau} d\tau b_{c\sigma}$, $\forall \sigma \in \mathbb{K}$ with $T = 0.5 [s]$.

As analyzed in [14], it is straightforward that the equilibrium point of the three level DC/DC multicellular converter is $x_d = [V_{dc}/3 \quad 2V_{dc}/3 \quad I_{ref}]$, where $I_{ref} \in [0, I_{\max}]$ and $I_{\max} = V_{dc}/R$. Therefore, φ_σ can be obtained as

$$\begin{aligned} \varphi_1 &= \begin{bmatrix} 0 \\ 0 \\ -2.594 \end{bmatrix}, \varphi_2 = \begin{bmatrix} -0.0233 \\ 0 \\ -1.7301 \end{bmatrix}, \varphi_3 = \begin{bmatrix} 0.0233 \\ -0.0233 \\ -1.7308 \end{bmatrix}, \varphi_4 = \begin{bmatrix} 0 \\ -0.0304 \\ -0.8656 \end{bmatrix}, \\ \varphi_5 &= \begin{bmatrix} 0 \\ 0.0068 \\ 0.9865 \end{bmatrix}, \varphi_6 = \begin{bmatrix} -0.0139 \\ 0.0139 \\ 1.8506 \end{bmatrix}, \varphi_7 = \begin{bmatrix} 0.0043 \\ 0.0048 \\ 1.8510 \end{bmatrix}, \varphi_8 = \begin{bmatrix} 0 \\ 0 \\ 2.7160 \end{bmatrix}. \end{aligned}$$

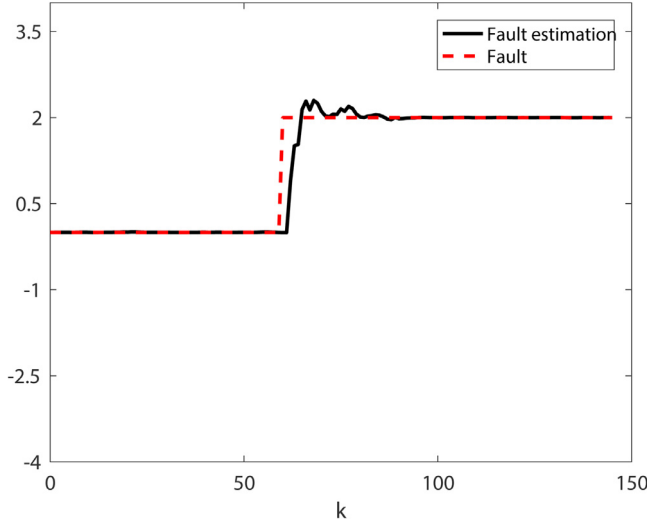


Fig. 2. Fault $f_1(k)$ and its estimate.

By solving the conditions derived in [Theorem 3](#), the fault estimator gains matrices can be calculated as follows

$$\begin{aligned}
 L_1 &= \begin{bmatrix} 1.0981 & 0.0457 & -0.0872 \\ 0.0362 & 1.0317 & -0.0054 \\ -0.0348 & 0.0714 & 0.2046 \end{bmatrix}, \quad L_2 = \begin{bmatrix} 1.0538 & -0.0063 & 0.0869 \\ 0.0123 & 0.9320 & 0.0002 \\ -0.2297 & 0.1235 & 0.6883 \end{bmatrix}, \\
 L_3 &= \begin{bmatrix} 0.9220 & 0.0450 & 0.1906 \\ 0.0528 & 1.0222 & -0.0359 \\ -0.1430 & 0.1310 & 0.2518 \end{bmatrix}, \quad L_4 = \begin{bmatrix} 0.7590 & -0.0245 & 0.0426 \\ -0.0171 & 1.0019 & -0.0441 \\ -0.3029 & 0.0358 & 0.2594 \end{bmatrix}, \\
 L_5 &= \begin{bmatrix} 0.9103 & -0.1228 & 0.1521 \\ -0.1597 & 0.6681 & -0.0714 \\ 0.0066 & -0.2542 & 0.3479 \end{bmatrix}, \quad L_6 = \begin{bmatrix} 2.1707 & 0.4227 & -0.6811 \\ -0.0068 & 0.8535 & -0.2371 \\ 0.2252 & -0.3511 & -0.0223 \end{bmatrix}, \\
 L_7 &= \begin{bmatrix} 0.9549 & -0.1599 & 0.1673 \\ -0.1601 & 0.1628 & -0.0117 \\ -0.0980 & 0.1539 & -0.0294 \end{bmatrix}, \quad L_8 = \begin{bmatrix} 1.2870 & 0.0480 & 0.0734 \\ 0.0314 & 1.0316 & -0.0519 \\ 0.1997 & -0.2165 & 0.0928 \end{bmatrix}, \\
 F_1 &= \begin{bmatrix} -0.1059 \\ 0.0120 \\ 0.1410 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 0.1698 \\ -0.0234 \\ -0.1704 \end{bmatrix}, \quad F_3 = \begin{bmatrix} -0.0985 \\ -0.0055 \\ 0.1695 \end{bmatrix}, \quad F_4 = \begin{bmatrix} 0.0965 \\ -0.0017 \\ -0.1303 \end{bmatrix}, \\
 F_5 &= \begin{bmatrix} -0.1255 \\ -0.0633 \\ 0.0352 \end{bmatrix}, \quad F_6 = \begin{bmatrix} -0.1214 \\ 0.1807 \\ 0.0214 \end{bmatrix}, \quad F_7 = \begin{bmatrix} -0.0314 \\ -0.0474 \\ 0.0682 \end{bmatrix}, \quad F_8 = \begin{bmatrix} 0.0239 \\ 0.1175 \\ -0.0612 \end{bmatrix}.
 \end{aligned}$$

with the H_∞ performance level $\gamma = 1.57$.

The disturbance signal in the multicellular DC-DC converter system is taken as $d(k) = \sin(0.5k)/k$. Firstly, we consider the fault signal as a constant value fault,

$$f_1(k) = \begin{cases} 0, & \text{otherwise} \\ 1.5, & k \in [60, 148] \end{cases}$$

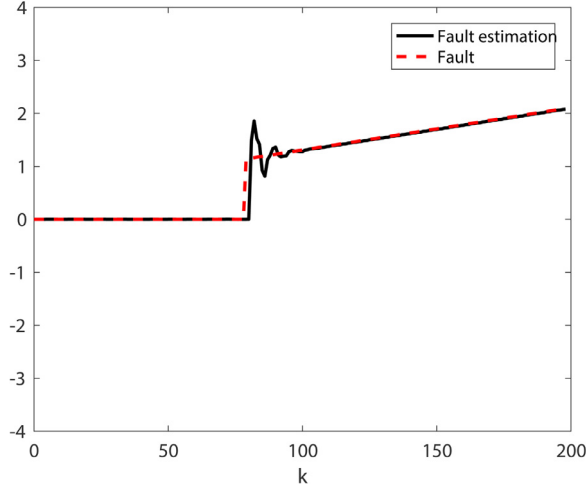


Fig. 3. Fault $f_2(k)$ and its estimate.

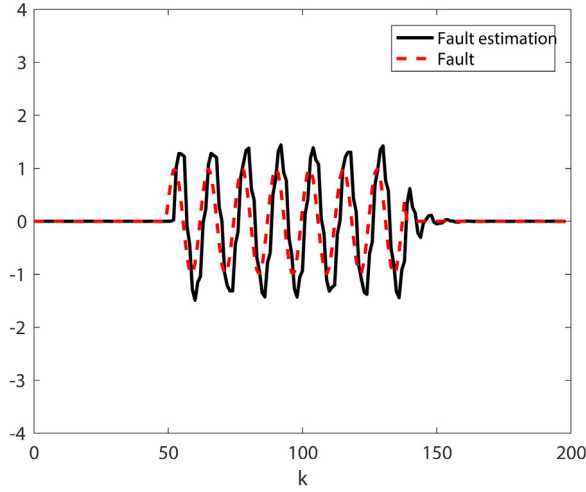


Fig. 4. Fault $f_3(k)$ and its estimate.

The fault estimation effect for fault signal $f_1(k)$ is shown in Fig. 2. When the fault signal is a time-varying signal, respectively as expressed in the following:

$$f_2(k) = \begin{cases} 0, & \text{otherwise} \\ 0.2 + 0.004k, & k \in [60, 148] \end{cases}$$

$$f_3(k) = \begin{cases} \sin(0.5 * k), & k \in [50, 140] \\ 0 & \text{otherwise} \end{cases}$$

At this time, Figs. 3 and 4 describe the fault estimation effect of fault signals $f_2(k)$ and $f_3(k)$, respectively.

From Figs. 2 to 4, it can be seen that no matter the fault signal is constant or time-varying, the fault estimation approach proposed in this paper can quickly and accurately realize the desired fault estimation.

6. Conclusions

In this paper, the observer-based fault estimation issue has been investigated for a class of discrete-time switched affine systems. The fault estimator design based on state observer has been converted into the target solution problem under the H_∞ performance index. In terms of Lyapunov Metzler inequality approach, the stability condition of augmented error system has been obtained by using the min-type Lyapunov function technique. By minimizing the volume of the ellipsoid containing the set of attraction, the fault estimator design has been realized which ensures the existence of numerical solutions in the form of linear matrix inequalities. Finally, an application to the three level multicellular DC-DC converter has been presented to verify the effectiveness and practicability of the proposed fault estimation scheme.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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